

Recap

Communications between clients and web servers are based on message exchanges

Clients send requests using HTTP request messages

Web servers provide responses using HTTP response messages

Messages include control data, a variable number of headers and an optional body

Control data in HTTP request messages specifies the action to be performed (e.g., download, upload) and the URI of the involved object

Control data in HTTP response messages defines the outcome of the request

Little differences in the syntax across HTTP versions (e.g., pseudo-headers)

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Status codes of HTTP response messages

The status code included in the status line refers to the outcome (i.e., success or failure) of the HTTP request

Codes are string of three digits subdivided in five groups:

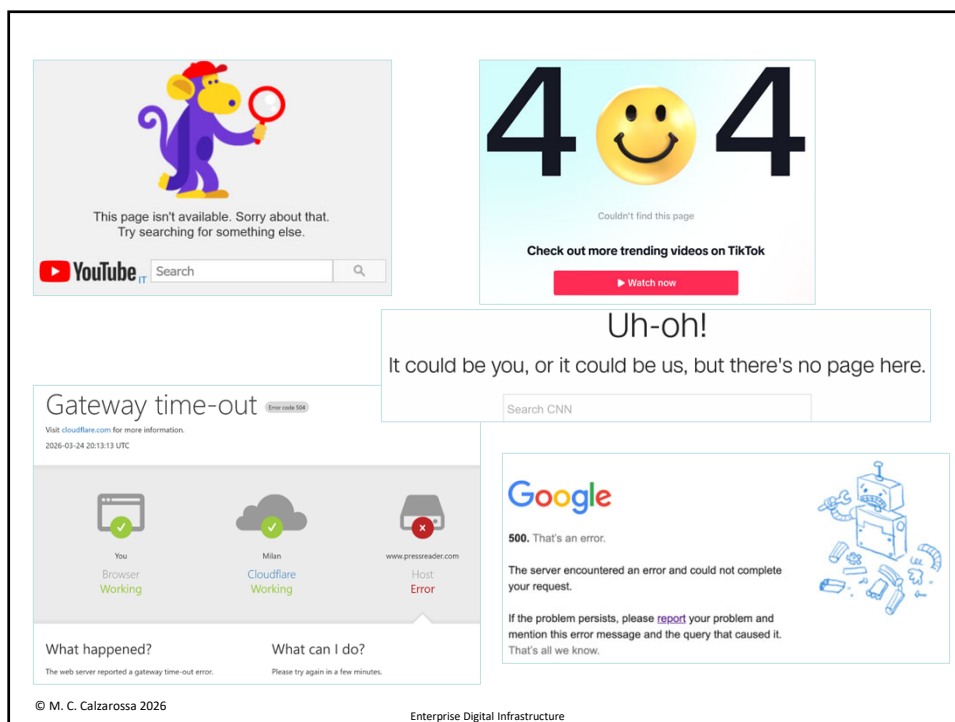
- 1XX: Informational
acknowledgement
- 2XX: Success
request received, understood & accepted
- 3XX: Redirection
additional client actions necessary to complete the request
- 4XX: Client error
request not processed due to a client error
- 5XX: Server error
request not processed due to a server error

Examples: **200** OK, **301** Moved permanently,
 404 Not found, **500** Internal server error

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Headers of HTTP request/response messages

Header fields are designed to exchange directives between clients and servers about their HTTP requests and responses

No big differences across HTTP versions (except for pseudo-header fields)

Clients use header fields to specify or advertise their preferences and configuration parameters; servers use header fields to specify how clients should handle the response

Standard format (case insensitive):

Field_name: Field_value

General and specific header fields of request/response messages

New header fields introduced to implement new features (*might be ignored*)

Browsers do not show request or response headers

HttpRequestHandler API provides actions to manipulate requests and responses

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Examples of request header lines

```
GET / HTTP/1.1
Host: www.unipv.it
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:149.0) Gecko/20100101 Firefox/149.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.9
Accept-Encoding: gzip, deflate, br, zstd
Connection: keep-alive
Upgrade-Insecure-Requests: 1
Sec-Fetch-Dest: document
Sec-Fetch-Mode: navigate
Sec-Fetch-Site: none
Priority: u=0, i
```

Mandatory (virtual hosting, intermediaries, proxies)

can be forged

"Fetch Metadata Request Headers" [W3C Working Draft, 1 April 2025]

Fetch metadata request header fields are a group of headers that give web servers information about the context in which the request is being made (e.g., navigation between documents)

Servers will be able to decide whether to allow or deny the request (defence mechanisms)

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Other examples of request header lines

<pre> :authority :method :path :scheme Accept Accept-Encoding Accept-Language Cache-Control Pragma Priority Sec-Ch-Ua Sec-Ch-Ua-Mobile Sec-Ch-Ua-Platform Sec-Fetch-Dest Sec-Fetch-Mode Sec-Fetch-Site Sec-Fetch-User Upgrade-Insecure-Requests User-Agent </pre>	<pre> portale.unipv.it GET / https text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7 gzip, deflate, br, zstd en-US,en;q=0.9,it;q=0.8,en-GB;q=0.7 no-cache no-cache u=0, i "Microsoft Edge";v="147", "Not.A/Brand";v="8", "Chromium";v="147" ?0 "Windows" document navigate none ?1 1 Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Edg/147.0.0.0 </pre>	Equivalent to Host: in HTTP/1.1 "User-Agent Client Hints" [W3C Draft Community Group Report, 10 February 2026]
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Client hints

Client hints are aimed at enforcing a model where the server must ask the browser for a set of data about the client (the hints) and the browser applies its own policies or user configuration to determine what data is returned

Access is managed in an explicit and auditable fashion rather than exposing all the User-Agent information by default

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Examples of response header lines

```
HTTP/1.1 301 Moved Permanently
Date: Sun, 19 Apr 2026 16:44:11 GMT
→ Server: Apache/2.4.38 (Debian)
→ Location: https://portale.unipv.it/
Content-Length: 312
Keep-Alive: timeout=5, max=100
→ Connection: Keep-Alive
Content-Type: text/html; charset=iso-8859-1
```

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